Emergency Tool-Use Safety Audit

Inspect a simulated engineering workspace, identify hazards, and produce a corrected setup plan.

Beginner	45-60 minutes	Practice & Review

Engineering Context

A small aerospace prototyping team is preparing for a rapid repair task. Before work begins, the team needs an independent safety review of tools, PPE, cords, materials, and traffic flow.

What You Will Practice

- Recognize common FabLab and shop hazards
- Match PPE to the task
- Communicate corrective actions clearly
- Apply stop-work authority responsibly

What You Need

- Instructor-provided workstation or safety photo
- Safety checklist
- Engineering notebook
- PPE examples

Student Instructions

1. Review the assigned workstation without touching anything.
2. Record at least eight observations: safe practices, hazards, or missing controls.
3. Classify each hazard by severity: low, medium, or high.
4. For every hazard, write one corrective action and who should complete it.
5. Create a revised workstation sketch showing safe tool placement, material flow, and walking space.
6. Perform a second inspection after corrections are made.
7. Write a brief go/no-go recommendation for starting work.

Engineering Requirements

- Identify at least five legitimate hazards or risk controls
- Include PPE, housekeeping, electrical, cutting/tool, and movement considerations
- Every hazard must have a specific corrective action
- Final recommendation must be supported by evidence

Constraints & Safety

- Do not operate tools during the audit
- Do not stage unsafe conditions for a photograph
- Follow instructor directions for handling PPE and equipment

What You Submit

- Completed safety audit
- Annotated workstation sketch
- Corrective-action table
- Final go/no-go recommendation
- Notebook reflection

Reflection Questions

1. Which hazard had the highest potential consequence?
2. Which correction was easiest to implement?
3. How can a safe workspace also improve efficiency?
4. When should a student stop work and ask for help?

Engineering Tip Separate the probability of an incident from the severity of its outcome when ranking risk.	**Common Mistake** Listing “be careful” is not a corrective action. State what must physically or procedurally change.
Stretch Goal Create a 60-second safety briefing that a team leader could deliver before the task begins.	**Career Connection** Aerospace technicians, test engineers, and manufacturing engineers complete pre-task hazard reviews before high-risk work.